

Towards Long-Term Pig Re-Identification: Investigating Human ReID Methods in Pigs

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Abstract

In precision livestock farming, early intervention at the individual level can improve health and welfare outcomes. However, automated long-term visual monitoring of individual pigs is challenging due to their lack of distinctive visual features and the scarcity of suitable public datasets. Existing research on this topic is limited to short time periods, of minutes or hours. Yet, to have a commercial impact, accurate monitoring over days to weeks is required. Hence, we study long-term pig re-identification, aiming to recognise individual pigs across multiple weeks despite variation in markings, weight gain, and growth. We evaluate existing human re-identification methods for this task using a newly released long-term pig tracking dataset. Our results show that while existing methods excel at same-day re-identification, there is a sharp drop-off in performance for cross-week scenarios, with many methods approaching chance performance by around six weeks. We also find that standard classification loss generally outperforms specialised re-identification loss functions, such as triplet loss, in the long-term setting.

Keywords: Long-Term Re-Identification, Pigs, Precision Livestock Farming

1 Introduction

With the advancement of precision livestock farming (PLF), it is now possible to monitor animals at the individual level rather than at the group level, allowing farmers to better understand an animal's behaviour, meet their nutritional needs, and identify health issues earlier for intervention. For instance, a pig with reduced feed intake can be identified for further checks to help the pig recover, prevent the spread of potential disease, and minimise financial loss. At the same time, farms are becoming larger whilst labour input per animal is decreasing, making manual individual observation impractical. This raises the problem of how to identify and track each individual pig over long time periods. While ear tags can be used in some circumstances, they are not practical for all scenarios [1]. To address the challenge of long-term non-invasive individual monitoring, computer vision-based multi-object tracking (MOT) algorithms have been explored [2]. However, trackers rely on continuous video input and accurate ID propagation between frames, which may provide a solution for short-term monitoring, but in real-world settings, errors accumulate due to factors such as occlusions and fast movements, leading to tracking failure [3]. A complementary approach is Re-Identification (ReID), which aims to match individuals over long time periods based purely on visual appearance in single images [4].

Re-identification methods were originally developed for matching people moving between cameras; adapting them to pigs poses several challenges. In contrast to human ReID, pig ReID is typically a single-camera task, and humans often wear distinctive clothing, but pigs tend to be visually similar, lacking clear distinguishing visual features [4]. In addition, factors such as illumination, viewing angle, skin stains, dirt, and the pig's growth introduce additional unwanted sources of appearance variation (Figure 1). Existing pig ReID benchmarks [4] are limited to short time periods of several minutes or hours and explore only a narrow range of

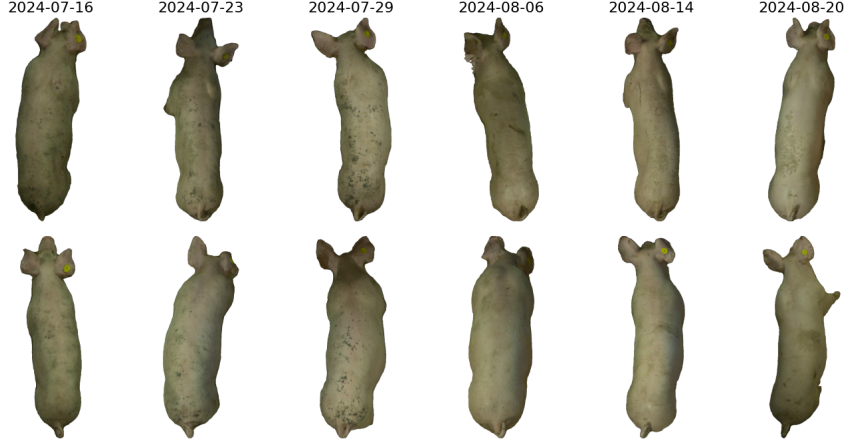


Figure 1: **Appearance variation of pigs over time. Each row represents an individual pig.** Due to the lack of distinct features, as well as minor drift of the camera angle, illuminations, stains and pigs’ growth, identification is difficult. Images extracted from [5]. The images were pre-segmented in the original dataset.

ReID training strategies, leaving long-term performance uncharacterised. Given these challenges, it remains an open question whether specialised ReID methods, mostly designed for humans or visually distinctive animals, remain effective for pigs, especially in long-term scenarios.

In this work, we leverage a recently released dataset for long-term, cross-day pig ReID [5], which provides longitudinal video recordings spanning more than one month on average and globally consistent pig identity labels across different video recording sessions. In the dataset, the pig images for ReID are pre-cropped and segmented to remove the interference of the pen background. We aim to understand how a wide range of ReID methods perform over days or weeks, extending the existing literature that largely focuses on shorter time periods. To achieve this goal, we evaluate the performance of multiple existing state-of-the-art re-identification methods over long time spans. To the best of our knowledge, this is the first time such analysis has been carried out. Hence, our work provides the community with a basis for future research on long-term pig re-identification.

2 Related Work

2.1 Existing Pig Tracking and Re-Identification Datasets

To satisfy the needs for long-term ReID, a dataset should provide i) high-quality ID annotations, such as globally consistent identity annotations across recording sessions to give indications whether a pig is the same pig in another video recording; ii) longer tracking spans of days, weeks, or months, instead of only a few minutes or hours; iii) the pigs in the dataset should not contain additional visible marks that can be exploited by the models. While several existing tracking-oriented datasets have been repurposed for pig ReID [6, 7], neither fully satisfies the requirements for long-term evaluation of pig re-identification.

The PigTrack dataset [6] contains dense labelling of each frame for pigs in diverse pen environments. However, the average monitoring span lasts only 30 seconds, which is useful for short-term tracking but less suitable for long-term ReID. In addition, we observed that some videos in the PigTrack dataset show clearly visible marks painted on the pigs’ backs, which the model can easily exploit instead of learning the pigs’ appearance characteristics. OinkTrack [7] extends the longest single-sequence monitoring time to 1 hour, emphasising the combined difficulties of occlusions and illumination changes during tracking and ReID. However, the dataset was sampled from only 2 pig pens, limiting its diversity, and the paper does not explicitly clarify whether its 574 reported identities are consistent across video sequences or only within each sequence, limiting the dataset’s applicability to ReID across recordings.

Bernotas *et al.* have recently introduced a longitudinal dataset for pig tracking and re-identification, target-

ing longer-term identity continuity [5]. It contains 20 monitoring groups, each corresponding to a fixed set of pigs housed together in a single pen. The dataset meets our key requirements: globally consistent ID labels, a longer monitoring span of 37.2 days on average (± 14.1 days, s.d.), and the absence of visible artificial marks on pigs. Instead of continuously recording, they captured videos on selected days, reducing the data volume for long-term coverage. To provide cross-video-consistent and long-term pig ID labelling, they made parallel recordings in visible and UV light. They then created unique sunscreen patterns on the backs of the pigs, visible only to UV cameras, not to the RGB camera used to capture the images in the dataset. Therefore, the ReID models must rely on the pigs' intrinsic appearance features, mimicking a realistic farm setting. Although the current public release does not contain the full original dataset due to ongoing curation, with only a small number of recordings missing, it remains suitable for our long-term evaluation of ReID performance. We note that the paper describing this dataset evaluates only multi-object tracking methods [5]. Hence, to our knowledge, we are the first to evaluate the performance of re-identification methods on this dataset.

2.2 Re-Identification Loss Functions

For ReID in the domains of pedestrians, wildlife and cattle livestock, the common loss functions used include ID losses, triplet losses, and contrastive losses [8, 9].

ID losses formulate ReID as a classification problem, typically optimised by the Cross-Entropy (CE) or a margin-based loss such as ArcFace [10]. ArcFace imposes an angular margin between the normalised feature embedding and the normalised class weight vector, encouraging intra-ID compactness and inter-ID separability. ID losses require a certain number of samples per identity to achieve optimal training performance, which can be problematic for short video clips. Testing on unseen identities is considered a disadvantage, such as unseen pigs in the test split, since the model is not explicitly optimised to separate them. Building on the ID loss framework, TAUDL extends cross-entropy for unsupervised cross-camera pedestrian ReID [11], enabling learning without clean identity labels or cross-camera annotations. However, it can also be used in supervised single-camera scenarios by seeing each individual pig group as a virtual camera and using ground-truth ID labels.

Triplet losses are metric learning methods that do not directly optimise the ID labels, but guide the appearance embedding to be closer together for positive samples and farther apart for negative samples. They do not rely on a predefined set of labels during training, and are more relaxed for unseen labels. They also directly optimise the image embedding used to measure image similarity. They are often combined with ID loss in person ReID models, such as BoT [12]. The mining strategy significantly affects the model training for triplet losses, namely how to select positive and negative pairs. For instance, hard-mining selects the hardest examples in training batches, i.e., the closest negative and the farthest positive example [13].

Contrastive losses such as the supervised contrastive loss (SupCon) [14] augment each input image twice to manually generate diverse domain differences. At each step, the loss function selects an image as an "anchor" and treats other augmented views with the same ID label as positive views. It pulls the distance between the anchor and the positive views closer and compares it against all other views except the anchor. Compared to ID losses and triplet losses, since it augments each image twice, it provides greater robustness to domain variance, but it doubles the training batch size, requiring more computing power.

3 Long-Term Pig Re-Identification

We define long-term pig re-identification as the problem of recognising an individual pig over long time periods, during which changes may occur in skin/coat markings, weight gain, and growth. We assume that pigs do not have unique artificial identifiers, such as ear tags, so that re-identification must be performed solely on visual appearance (Figure 1). The core task is to match a query image to a gallery captured days or weeks earlier and retrieve all gallery images of the same pig. Both the query and gallery may contain multiple images per individual. In commercial settings, this can be formulated as closed-set matching because the pigs within a pen are typically fixed. Accordingly, each query identity is present in the gallery, while all validation and test identities remain unseen during training. Thus, the models must generalise to unseen pigs.

Split	# Groups	# Pigs	# Pigs Per Group	Recording Interval (Days)	Recording Times
Train	12	102	8.50 ± 4.96	4.02 ± 3.03	8.85 ± 4.35
Validation	2	12	6.00 ± 0.00	7.00 ± 0.78	7.50 ± 0.50
Test	2	12	6.00 ± 0.00	7.00 ± 0.78	7.50 ± 0.50

Table 1: Statistics of our train/validation/test splits of [5].

3.1 Re-Identification Model

We use an ImageNet-pretrained ResNet50 as the feature-extraction backbone in all ReID experiments. Feature maps from the final convolutional layer are globally average-pooled and flattened to produce pig appearance descriptors. Horizontal flipping is used for data augmentation. No explicit head-to-tail orientation normalisation is applied, and the backbone is not assumed to be strictly rotation-invariant. We fine-tune the backbone with Cross-entropy (CE), ArcFace [10], Triplet loss [13], Triplet loss + CE combined (CE+Triplet) [12], SupCon [14], TAUDL and TAUDL with global header (TAUDL_G) [11].

CE, ArcFace, Triplet, CE+Triplet and SupCon Training batches are constructed by selecting P pigs with K images per pig. The backbone outputs are fed into a BNNeck [12] to normalise embeddings. During training, the cross-entropy and arcface losses are computed on features after BNNeck, while the triplet loss is computed on features before BNNeck. During evaluation, features are extracted after BNNeck. All ID losses use global classification, computed as a softmax over all pig identities in the dataset. For distance calculation, the CE model does not need to calculate any distance; ArcFace uses the product of the normalised output-layer classification weight vector and the normalised output embedding vector to compute the angle; Triplet and the CE+Triplet variant use Euclidean distance, and SupCon uses cosine similarity for the difference between the anchor and samples. We use batch-hard-mining for Triplet and CE+Triplet, and no mining for SupCon.

TAUDL Since pigs in different groups do not overlap, we treat each pig group as a separate camera, disable cross-camera matching, and directly use the dataset’s ground-truth labels during training. We adopt TAUDL’s camera-wise sampler, which in our case samples C groups and I instance images per pig. Backbone output after average pooling and ReLU activation is fed to separate classification heads for each pig group, with cross-entropy computed within each head. As a result, the model does not explicitly learn the discrimination across different pig groups. This is intentional and designed to simulate a setting in which global ID labels are unavailable across video recordings. We also evaluate a TAUDL variant with a global classification head, TAUDL_G, which quantifies the value of cross-session global ID labels.

4 Experiments

4.1 Dataset

We use a subset of the recently released Large-Scale Longitudinal Pig ReID dataset (PigReID) for our experiments [5]. In the subset, the statistics for pig groups 1 to 13, 17, 18, and 20 match those of the full dataset; groups 14, 16, 19, and 21–22 are missing; group 18 is missing one pig identity. Full details are reported in Appendix A. We manually split the dataset into train, validation, and test sets at the group level, ensuring no overlap between pigs and pens across the splits. Both the validation and test sets are selected to include 2 pig groups of 6 pigs each to ensure balanced evaluation. The time interval between each pig’s recording sessions in both sets is approximately 7 days (one week). The remaining 12 groups are used for model training. Statistics of the splits are shown in Table 1 and Appendix A.

4.2 Evaluation Protocol

To evaluate ReID performance across temporal separations between recordings, we organise evaluation pairs into *session-gap buckets*. Each pig group c is recorded n_c times during the monitoring period, producing

non-overlapping, discontinuous video sessions $v_{c,1}, v_{c,2}, \dots, v_{c,n_c}$ sorted by date in ascending order, where each session corresponds to a single video that contains images of all pigs in group c . For each group, a pair of sessions $(v_{c,i}, v_{c,j})$ with $i \leq j$ is assigned to the session-gap bucket $b = j - i$, where $v_{c,i}$ provides the query images and $v_{c,j}$ provides the gallery images. In the test and validation setting described in Section 4.1, each session is recorded 7.00 ± 0.78 days apart for each pig group, bucket 0 corresponds to a 0-day gap (same session), bucket 1 to an approximately 7-day gap, and bucket 2 to an approximately 14-day gap, etc. For bucket 0 (i.e., $i = j$), where query and gallery come from the same session, to avoid overlaps, the gallery images are sampled from the first 30% of the session frames, the query images are sampled from the last 30% of the session frames, and the middle 40% of the session frames are discarded.

For one evaluation, we randomly sample one session pair from a session-gap bucket b per pig, and draw N_q query images from $v_{c,j}$ and N_g gallery images from $v_{c,i}$. When multiple images are used, the embeddings are averaged for matching. We evaluate re-identification performance across session-gaps using the Cumulative Matching Characteristic (CMC) Rank-1 Score, defined as the proportion of queries whose highest-ranked gallery descriptor matches the query pig’s identity, with larger values indicate better performance.

4.3 Experimental Settings

For all experiments we use input images of size 384x384 pixels and a batch size of 64. Due to the specificities of contrastive learning, the batch size for SupCon changes to 128 after augmentation. For non-TAUDL models, we sample 4 instances per pig per batch. For TAUDL models, we sample 8 pig groups and 4 instances per pig per batch. Hyperparameters, including training steps, are searched based on the average CMC Rank-1 across all session-gaps in the validation set, and the final performance is reported per session-gap in the test set, averaged over 10 repetitions. Optimal hyper-parameter settings can be found at Section B. All models were trained on the AMD MI300X GPU using the Adam optimiser and a cosine learning rate scheduler.

5 Results

Results in Figure 2 show the CMC Rank-1 performance and standard deviation of all re-identification models across different session-gaps ($g_{_}$) with the corresponding range of calendar days ($_{_}d$), and increasing numbers of query and gallery images per pig. Darker colours indicate better performance.

Long-term Re-Identification Across all models, the strongest performance occurs when query and gallery images are drawn from the same day, i.e., session-gap 0. With six pigs per pen in the validation and testing sets, the at-chance Rank-1 CMC is $\approx 16\%$. Hence, all models demonstrate performance well above chance at session-gap 0, with the best-performing models exceeding 85% across all query and gallery settings. For all models, performance generally decreases as the session-gap increases, and several are near chance level by around 6 weeks (42 days). These results underscore the challenge of long-term pig re-identification, where factors such as growth, appearance variation, and environmental changes may adversely affect model performance over long time periods. Averaged across all session-gaps, the best-performing models are Pure CE and CE+Triplet. This demonstrates that standard ID loss functions, such as cross-entropy, are effective for long-term pig ReID tasks. In contrast, Pure Triplet loss performs poorly across all settings on this dataset. Apart from the case of 1 query and 1 gallery image per pig, which may, however, be more strongly affected by chance factors. The TAUDL and TAUDL_G methods differ by around 6–7% in average performance, suggesting the value of global constant ID across videos, although maintaining such identity information across independent recording sessions may be challenging in commercial deployments.

Varying Query and Gallery Images Increasing the number of query and gallery images improves the performance of all methods, including over longer periods. We hypothesise that increasing the number of images per pig reduces visual noise by diminishing the influence of chance factors, such as occlusion or illumination, which may degrade individual-specific information in randomly selected images. However, simply increasing the number of images per pig is not enough to enable reliable long-term re-identification alone, as even with 5 query and 5 gallery images per pig, performance drops to near chance for many models by around 42 days.

CMC Rank-1 mean and std by session and calendar days (%)																													
1 Query 1 Gallery										3 Query 3 Gallery										5 Query 5 Gallery									
ArcFace mean	85.0	36.7	26.7	27.5	18.3	24.2	15.8	33.5		98.3	41.7	33.3	30.0	24.2	21.7	20.8	38.6		97.5	39.2	33.3	27.5	29.2	30.8	25.0	40.4			
std	6.2	12.5	11.7	12.4	11.7	10.8	6.9	10.3		3.3	15.8	11.2	13.5	11.5	10.0	11.3	11.0		3.8	11.2	11.2	9.9	6.7	11.8	10.5	9.3			
CE+Triplet mean	76.7	41.7	30.0	30.8	23.3	20.0	15.0	33.9		96.7	50.0	45.0	38.3	28.3	25.0	23.3	43.8		95.0	51.7	45.0	33.3	26.7	30.0	19.2	43.0			
std	12.2	11.2	15.9	11.2	9.7	10.0	9.7	11.4		4.1	9.9	13.5	13.0	15.5	11.2	11.7	11.3		5.5	15.3	10.7	7.5	7.3	9.3	8.4	9.1			
Pure CE mean	86.7	45.0	30.8	34.2	27.5	15.8	19.2	37.0		96.7	50.0	37.5	29.2	28.3	31.7	22.5	42.3		96.7	46.7	40.0	33.3	25.8	31.7	26.7	43.0			
std	9.3	11.9	14.9	16.9	15.4	9.5	7.5	12.2		4.1	12.9	11.3	10.7	11.9	10.4	10.6	10.3		5.5	8.5	11.1	5.3	7.9	13.8	12.8	9.3			
Pure Triplet mean	38.3	27.5	22.5	22.5	21.7	20.0	15.8	24.0		64.2	30.0	26.7	22.5	28.3	22.5	22.5	31.0		79.2	33.3	33.3	31.7	22.5	19.2	26.7	35.1			
std	7.6	13.5	10.6	7.5	17.2	10.0	12.0	11.2		11.2	11.3	11.1	9.2	7.6	9.2	8.4	9.7		11.9	7.5	15.4	9.0	11.8	10.6	7.3	10.5			
SupCon mean	54.2	41.7	37.5	26.7	32.5	22.5	21.7	33.8		84.2	45.0	40.0	30.0	29.2	30.0	21.7	40.0		85.8	44.2	49.2	36.7	31.7	30.8	22.5	43.0			
std	13.6	9.9	18.0	13.8	13.7	9.2	9.3	12.5		7.9	10.0	7.3	7.6	11.3	6.7	10.0	8.7		9.9	15.8	10.8	8.5	11.1	9.9	8.4	10.6			
TAUDL mean	66.7	39.2	28.3	26.7	26.7	23.3	18.3	32.7		86.7	42.5	25.0	35.0	28.3	17.5	20.8	36.5		92.5	45.8	33.3	30.8	25.8	15.8	15.8	37.1			
std	11.8	12.9	8.5	8.2	11.7	12.8	10.4	10.9		9.3	9.5	11.2	8.2	7.6	7.9	6.7	8.6		7.9	11.3	6.5	7.5	10.8	6.9	4.5	7.9			
TAUDL_G mean	65.0	30.0	35.8	30.8	32.5	24.2	26.7	35.0		87.5	45.8	38.3	35.8	30.0	27.5	37.5	43.2		94.2	50.0	45.8	38.3	24.2	25.8	27.5	43.7			
std	15.7	14.5	17.5	11.8	9.5	14.6	10.4	13.4		8.5	10.7	12.5	10.6	7.6	9.9	10.7	10.1		7.5	13.9	11.3	10.0	9.5	6.9	7.5	9.5			
	g0	g1	g2	g3	g4	g5	g6	AVG		g0	g1	g2	g3	g4	g5	g6	AVG		g0	g1	g2	g3	g4	g5	g6	AVG			
	0d	6-8d	12-16d	20-22d	28-29d	34-36d	41-43d	AVG		0d	6-8d	12-16d	20-22d	28-29d	34-36d	41-43d	AVG		0d	6-8d	12-16d	20-22d	28-29d	34-36d	41-43d	AVG			
	Session gap / calendar days																												

Figure 2: CMC Rank-1 Re-Identification performance across long time periods, using different numbers of images per pig for query and gallery. Darker colours represent better performance. For each method, the mean and standard deviation (std) over 10 runs are reported. Columns are labelled by session-gap g_* and corresponding calendar days $_d$.

6 Conclusion

In this paper, we have, for the first time, compared the long-term performance of different ReID methods in pigs, using a new public dataset. Most methods perform well on the same-day re-identification task, achieving at least 75% CMC Rank-1 except for Pure Triplet loss, demonstrating that human ReID methods are capable of performing short-term pig ReID tasks. At the same time, the performance of all evaluated methods degrades over longer time periods. Many methods approach chance-level performance at a gap of around 42 days between query and gallery images. The use of multiple query and gallery images per pig generally improves the performance of all the re-identification methods tried, but alone, this is not enough to enable practical long-term re-identification. The best-performing long-term re-identification methods make use of standard cross-entropy loss, with Pure CE and CE + Triplet losses achieving the best performance. We hope that our analysis will serve as a starting point for future research on long-term pig re-identification, helping to enable individual-level interventions for improved health and welfare outcomes.

7 Limitations and Future Work

Although we experimented with various ReID methods, we used ResNet backbone to extract embeddings. Future work may extend to a backbone based on the Vision Transformer (ViT) and orientation normalisation using a pig body keypoint estimation model to align images along the head-to-tail axis. Limited by the currently available dataset size, our evaluation set contains only 12 pigs, resulting in relatively large variance and limiting fine-grained statistical comparisons between methods. We expect this to be addressed once the full dataset becomes available.

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A Appendix: Data subset statistics

Split	Group	# Pigs	Recoding Interval (Days)	Max Interval (Days)	Min Interval (Days)	Recording Times
Train	G3	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Train	G4	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Train	G5	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Train	G6	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Train	G7	7	7.25 ± 0.83	8	6	5.00 ± 0.00
Train	G8	7	7.25 ± 0.83	8	6	5.00 ± 0.00
Train	G9	7	6.83 ± 0.90	8	6	7.00 ± 0.00
Train	G12	5	6.83 ± 0.90	8	6	7.00 ± 0.00
Train	G13	7	3.64 ± 1.15	6	2	12.00 ± 0.00
Train	G17	20	2.44 ± 2.38	7	1	4.50 ± 0.92
Train	G18	19	2.21 ± 2.77	11	1	14.47 ± 2.23
Train	G20	6	1.00 ± 0.00	1	1	18.00 ± 0.00
Validation	G1	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Validation	G10	6	6.83 ± 0.90	8	6	7.00 ± 0.00
Test	G2	6	7.14 ± 0.64	8	6	8.00 ± 0.00
Test	G11	6	6.83 ± 0.90	8	6	7.00 ± 0.00

Table 2: Statistics of the defined PigReID train/validation/test split. The observe interval is computed from adjacent recording dates of the same pig, and sampling times denote the number of recording dates per pig.

B Appendix: Loss Function Hyper-parameters

Loss name	Hyper-parameters
CE+Triplet	$id_loss_weight = 0.3$; $tri_loss_weight = 0.7$; $margin = 0.2$; $LR = 6e-6$; $steps = 16025$
ArcFace	$id_loss_scale = 48$; $id_loss_margin = 0.35$; $LR = 5e-6$; $steps = 25640$
SupCon	$temperature = 0.03$; $LR = 1e-5$; $steps = 6410$
TAUDL	$head = multi_head$; $LR = 6e-5$; $steps = 1000$
TAUDL_G	$head = single_head$; $LR = 6e-5$; $steps = 1000$
Pure Triplet	$id_loss_weight = 0.0$; $tri_loss_weight = 1.0$; $margin = 0.2$; $LR = 6e-6$; $steps = 16025$
Pure CE	$id_loss_weight = 1.0$; $tri_loss_weight = 0.0$; $LR = 6e-6$; $steps = 16025$

Table 3: Loss hyper-parameters. id_loss_weight and tri_loss_weight are the factors for ID loss and triplet loss, respectively, that contribute to the final loss; the two factors sum to 1. LR represents the learning rate.